

# Automating the installation of cables

## Los pollos hermanos

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**Abstract**—Cable and wire manufacturing is a high-volume sector with a stable, growing demand. This trend can make automation an interesting opportunity for manufacturers to reduce production costs and increase their profits.

This study focuses on the automation of the cable crimping process with a robot arm and the comparison of different algorithms for cable detection and grasping estimation. Three algorithms based on Convolutional Neural Networks (CNNs) and Hue, Saturation, Value (HSV) segmentation were compared against the State-of-the-art method, both in simulation and with a physical setup.

**Index Terms**— *Deformable Linear Object, Sim-To-Real, Cable Crimping*

### I. DESCRIPTION OF THE INDUSTRIAL CHALLENGE

The industrial challenge presented revolves around the detection and manipulation of Deformable linear objects-DLO. The detection and tracking of DLOs must be very precise because of these objects' elastic properties and shape. Moreover, the detection of the grasping point requires high precision, robustness to occlusions and lighting conditions. The automation of some processes in cable manufacturing has the potential to allow manufacturers to cut costs in their production, standardize their quality and increase volume. In addition to that, replacing manual labor in the cable crimping process can prevent injuries caused by the repetitive actuation of crimping machines and tools.

### II. STATE OF THE ART

The domain of DLO's tracking has undergone a significant paradigm shift over the last decade, evolving from rigid heuristic methods to data-driven deep learning, and finally to physics-aware systems. This evolution can be categorized into three distinct eras:

#### A. The Geometric Era: Classical Computer Vision (Pre-2015)

Early approaches to cable manipulation relied heavily on Classical Computer Vision techniques. These methods were "model-free" and generally heuristic-based, prioritizing computational efficiency over semantic understanding.

- 1) **Techniques:** Engineers utilized color space transformations (e.g., HSV or YCbCr thresholding to isolate specific colors Chen et al. [2015]) and edge detection algorithms (e.g., Canny in Chen et al. [2015] and Robinson compass masks in Parmar [2014]) to segment the cable from the background based on pixel intensity gradients.

- 2) **Limitations:** While fast, these algorithms were highly sensitive to environmental variables. Changes in illumination, background clutter, or shadows often caused tracking failures. Furthermore, they lacked 3D depth perception; if a cable looped over itself, the algorithm typically viewed it as a disconnected intersection rather than a continuous object (Parmar [2014]).

#### B. The Data-Driven Era: Deep Learning and CNNs (2015–2021)

The advent of Convolutional Neural Networks (CNNs) revolutionized DLO tracking by replacing manual rule-programming with learned feature extraction. Instead of defining how to find a cable, engineers trained models on large datasets of annotated cable images.

- 1) **Techniques:** CNN classify pixels individually as "cable" or "background," allowing for robust segmentation even in unstructured or "messy" environments (Caporali et al. [2022]), and specialized models (Pose Estimation) were developed to identify specific topological features, such as the cable tip, the grasping point, or the center of a loop, treating the cable similarly to a human skeleton.
- 2) **Limitations:** While excellent at detection, purely visual deep learning models often lack temporal consistency and can lead to physically impossible shape predictions when the cable is partially obscured (Xiang et al. [2023]).

#### C. The Physics-Aware Era: Hybrid Intelligence (2022–Present)

The current State of the Art (SOTA) acknowledges that vision alone is insufficient for high-precision, high-complexity tasks, particularly when the robot arm occludes the camera or when the cable enters a machine. The modern approach fuses AI perception with physics-based modeling and tactile feedback.

- 1) **Physics-Informed Neural Networks (PINNs):** These algorithms bridge the gap between vision and mechanics. By incorporating mathematical models of rod mechanics (such as Cosserat Rod Theory (Spillmann and Teschner [2007])), these systems can predict the cable's behavior. A critical advantage of this approach is robustness to occlusion. If the robot blocks the camera's

view, the physics model calculates where the hidden segments must be based on material stiffness and curvature continuity, ensuring the robot maintains state estimation.

- 2) **Sim-to-Real Transfer:** Current research utilizes hyper-realistic physics simulations to train agents Matas et al. [2018]. Robots perform the desired task thousands of times in simulation, can learn complex strategies (Sim2Real) that are then transferred to the physical robot.

### III. APPROACH

This work focused mainly on the detection and grasping point estimation of the cable. Two different algorithms based on color segmentation and CNNs were compared with an SOTA tracking algorithm. The reason for this is the lack of need for real time tracking for this particular task, where more time on the detection step is allowed to get a better estimate and the real-time requirements are relaxed.

The major drawback of these methods is generally the lighting conditions. For the case of color-based algorithms, this is particularly important, as shadows and reflections can reduce the quality of the segmentation and lead to poor performance. On the other hand, CNN-based methods are more computationally expensive and their color independence can backfire if some other DLOs are present in the scene, such as robot cables. An overview of the different algorithm characteristics can be found in Table III.

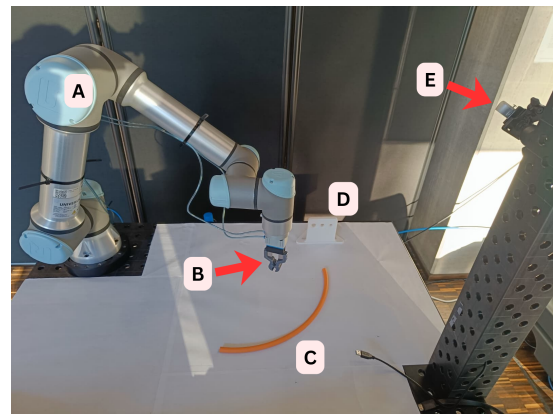
TABLE I: *Algorithms overview*

| Algorithm features        | Algorithms |         |          |
|---------------------------|------------|---------|----------|
|                           | HSV-Seg    | FastDLO | TrackDLO |
| Tracking                  | no         | no      | yes      |
| Color based               | yes        | no      | yes      |
| Robust to self-occlusions | no         | yes     | yes      |

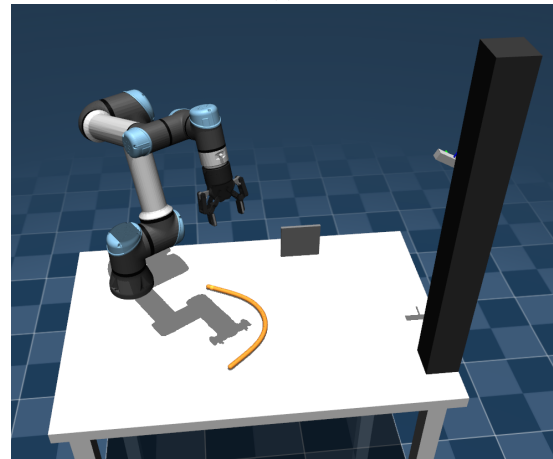
An experiment was conducted on the real setup, in which each algorithm was run 20 times on the same cable pose and 15 evenly spaced points along the cable were sampled. On the simulated setup, a similar experiment was conducted, comparing FastDLO and HSV-Seg algorithms' detections in increasing levels of noise.

### IV. EXPERIMENTAL PROTOTYPE

The hardware consists of a UR5e robot (Fig. 1), equipped with a custom-designed pneumatic gripper (Fig. 2), an orange cable, an Intel Realsense D415 3D camera and a wall mockup (Fig. 3). The pneumatic gripper has a maximum travel of 10 mm, which leaves only a little room for play during the grasping motion. To still be able to account for cable-tracking and calibration inaccuracies, the gripper was designed to have a bigger opening at the bottom with a slight chamfer to direct the cable into the gripper (Fig. 2 (c) and (d)). As the gripper fingers were designed for a cable with a diameter of 13 mm, in the open state, there is 3.4 mm of room for inaccuracies. Furthermore, the wall thickness of the tip of the gripper is kept



(a)



(b)

Fig. 1: (a) Experimental setup. A: UR5e robot, B: pneumatic gripper, C: orange cable, D: wall mockup with hole, E: Intel Realsense D415 3D camera (b) Simulation in MuJoCo of the whole robotic setup.

low, compared to the rest, to accommodate the cable lying on the ground.

To simulate a cable crimping machine, a wall was designed, which can be seen in Fig. 3, with three different hole sizes (diameters of 16 mm, 13 mm and 10 mm) to be able to test different cables. The cable is then autonomously inserted into a hole to see if the goal is achieved. To achieve the goal of a functional prototype, the UR5e robot autonomously picks up the cable and inserts it into a hole in the wall, the camera has to be calibrated to the robot. Meaning, accurately calculating the transformation between the camera coordinate frame and the robot's base coordinate frame. This is necessary because the tracked 3D points of the pose of the cable are given in reference to the camera's coordinate frame. However, for the robot to be able to accurately position the gripper over the cable, it has to know the exact coordinates in the robot's base frame. In the end, it involves a simple matrix-vector multiplication with the matrix being the homogeneous transformation matrix from camera frame to robot-base frame,

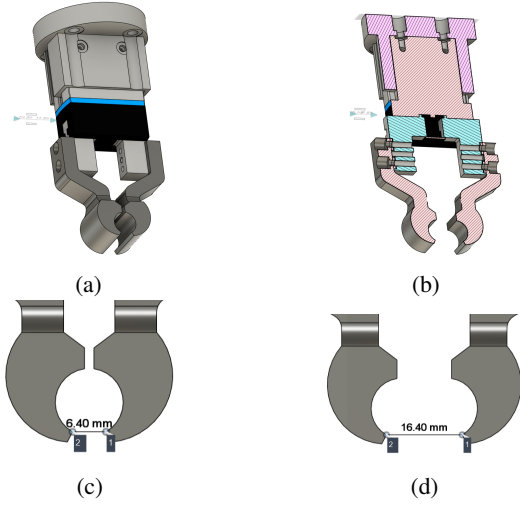


Fig. 2: Gripper design, optimized for a cable with a diameter of 13 mm. (a) Pneumatic gripper in closed state. (b) Pneumatic gripper in half-section view in open state. (c) Gripper close-up in closed state with a 6.4 mm opening gap. (d) Gripper close-up in opened state with a 16.4 mm opening gap.

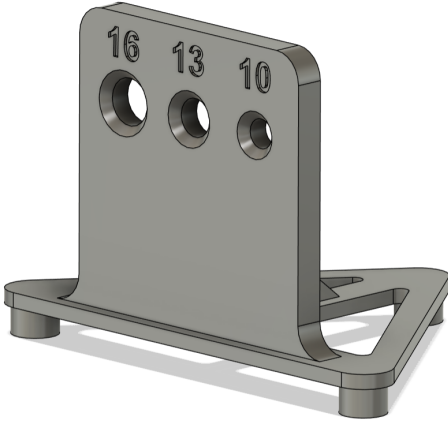


Fig. 3: Wall mockup for simulating a cable crimping machine. It has 3 holes of diameter 16 mm, 13 mm and 10 mm.

the vector is a 3D point in the camera's frame homogenized. To estimate the transformation matrix, the hand-to-eye calibration method is used. It works by observing a calibrated target from multiple known robot poses and computing the relative pose between the robot's end effector and the camera across all sampled observations. In this case, a ChArUco board was used (Fig. 4) with a grid size of  $14 \times 9$ , square side length of 20 mm and marker side length of 14 mm. With the board directly mounted on the robot's end effector, 20 different samples were recorded and ran the calibration algorithm.

For each sample  $i$ , the end effectors pose  $A_i$ , in the robot-base frame, and the ChArUco board pose  $B_i$ , in the camera frame, is known. Solving the linear equation system with equation 1 results in the homogeneous transformation  $X$

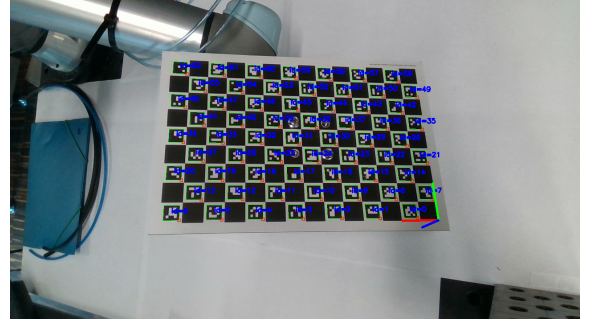


Fig. 4: ChArUco board tracked with OpenCV. Due to the known marker size and camera intrinsics, an approximate homogeneous transformation from a 2D image can be computed.

between the robot-base frame and camera frame.

$$A_i X = X B_i \quad (1)$$

The method used for efficiently solving this problem is described in Tsai and Lenz [1989] and implemented in the OpenCV library for Python (Bradski [2000]).

The next steps were to detect and track the cable. For that, the ROS2-Jazzy implementation of TrackDLO was used, which publishes an array of ordered 3D points, in camera space, that skeletonizes the cable. Two consecutive points are then transformed to robot space by multiplying them by the transformation matrix computed by hand-eye calibration. The relative vector between these two points is then used to calculate the wrist rotation of axis 6 of the robot, to correctly rotate the gripper to be aligned with the path of the cable. The robot then drives to the first transformed point in the cable, by approaching it 3 cm above the cable and then slowly moving down. The gripper closes and the robot moves to a predefined pose of a 16mm hole in the wall, which resembles the cable crimping machine and slowly inserts the tip of the cable.

## V. SIMULATION

The cable tracking algorithms were also implemented and rigorously tested within a high-fidelity MuJoCo simulation environment. Utilizing a simulation parallel to the physical setup offers significant advantages for robotic manipulation tasks involving DLOs.

### A. Key Benefits of Simulation

- **Movement Verification:** Simulation allows for the verification and validation of robot movements and trajectories before execution on the physical hardware, greatly reducing the risk of errors or damage.
- **Performance Benchmarking:** The environment facilitates checking the **accuracy and robustness** of the implemented algorithms (HSV-Seg, FastDLO, and TrackDLO) under a variety of controlled conditions, such as systematically **increasing image noise**.
- **Advanced Control Development (Sim-to-Real):** Simulation serves as a critical platform for training advanced

robot control algorithms, such as **Reinforcement Learning (RL)**. The ability to perform thousands of attempts in the simulated environment allows agents to learn complex, optimal strategies (the Sim-to-Real transfer paradigm) that can then be applied to the physical robot.

### B. Data Acquisition in Simulation

To assess algorithm performance and provide the necessary input for DLO tracking, the MuJoCo environment provided four core data modalities. (Fig. 5)

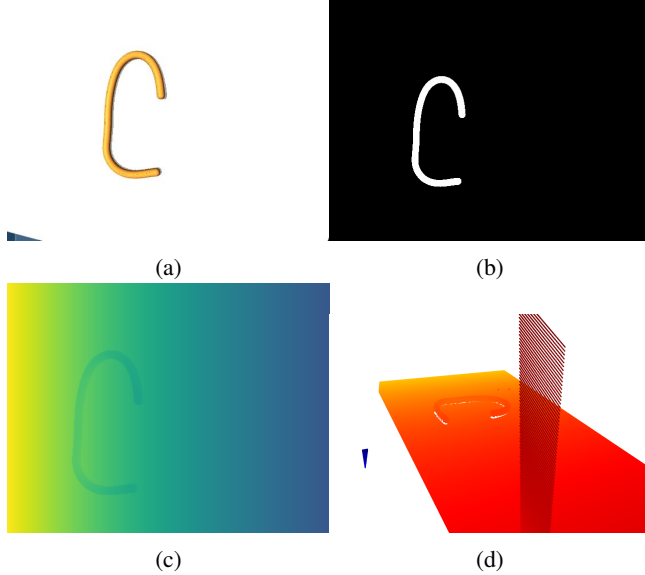


Fig. 5: Data obtained from MuJoCo simulation: (a) RGB, (b) Segmentation, (c) Depth, and (d) Point cloud.

To evaluate algorithmic robustness, real-world environmental degradations were simulated using a combined noise parameter  $N \in [0, 50]$ . This parameter controls three distinct models: Gaussian noise ( $\sigma = N$ ) to simulate sensor variance; salt-and-pepper noise ( $d \approx N/255$ ) to mimic dust and artifacts; and speckle noise ( $f = N/100$ ) to replicate coherent light interference. By increasing  $N$  to 50, the "worst-case" industrial conditions were simulated, such as high-gain sensor noise and physical particulates to identify the breaking point of the cable tracking algorithms. (Fig. 6)

### C. Algorithm Implementation for Cable Pose Estimation

The goal of the segmentation phase is to generate a precise binary mask of the cable from the input RGB image, a necessary precursor to pose estimation. Two contrasting approaches were implemented:

- **FastDLO (Convolutional Neural Network-based)** The FastDLO algorithm takes the input RGB image and generates a segmentation mask using a CNN. This deep learning approach relies on features learned from a large dataset of annotated cable images to classify pixels as

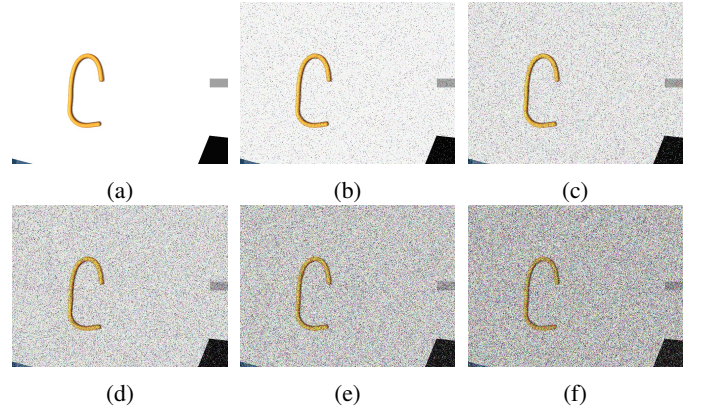


Fig. 6: RGB images with different noise levels. a) No noise. b) Noise level 10. c) Noise level 20. d) Noise level 30. e) Noise level 40. f) Noise level 50.

either "cable" or "background", offering robust segmentation even in environments with complex textures or varying lighting.

- **Input:** RGB image.
- **Method:** CNN inference.
- **Output:** Binary segmentation mask of the cable.
- **HSV Segmentation (Classical Computer Vision-based)** The HSV Segmentation algorithm is a classical computer vision approach that also generates a segmentation mask from the RGB image. Unlike FastDLO, it does not require a deep learning model. Instead, it utilizes color information by converting the RGB image to the Hue, Saturation, Value (HSV) color space and applying pre-defined thresholding to isolate the specific color of the cable (e.g., orange) from the background.
  - **Input:** RGB image.
  - **Method:** Color space transformation and HSV thresholding.
  - **Output:** Binary segmentation mask of the cable.

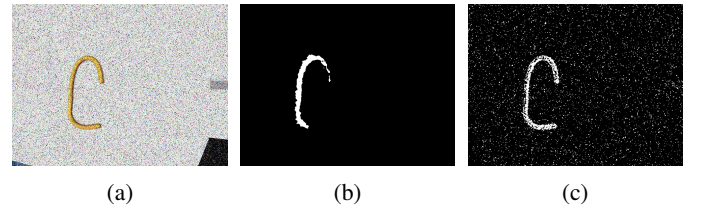


Fig. 7: Binary masks of FastDLO and HSV segmentation. a) Input Image (Noise=30). b) FastDLO mask. c) HSV segmentation mask.

- **Skeletonization and Node Extraction** Both the FastDLO and HSV segmentation masks require a subsequent processing step—skeletonization—to convert the 2D cable area into a sequence of ordered 3D points (nodes) that define the cable's centerline. The Skeletonization Algorithm consists of the following four sequential steps:



- 1) **Preprocessing and Noise Removal:** The binary mask is cleaned using morphological operations: Opening removes specks of noise, while Closing fills minor holes and gaps within the cable area. Finally, only the largest connected component is retained to eliminate any large, disconnected noise blobs.
- 2) **Skeletonization:** The cleaned mask is reduced to a single-pixel-wide line—the skeleton—using the `skimage.morphology.skeletonize` function Pedregosa et al. [2011].
- 3) **Graph Traversal and Path Ordering:** The skeleton pixels are converted into a NetworkX graph. The main centerline of the cable is identified by finding the shortest path between the two furthest-apart endpoints of the graph, which effectively orders the points and prunes small skeletonization artifacts (branches).
- 4) **Resampling and 3D Extraction:** Linear interpolation is applied to the ordered path to generate a specified number of equidistantly spaced nodes (e.g., 15 nodes). The final 3D pose is obtained by using the 2D pixel coordinates of these nodes to look up the corresponding depth value from the simulation's depth information.

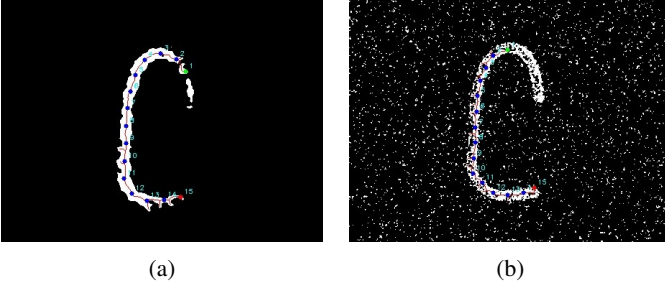


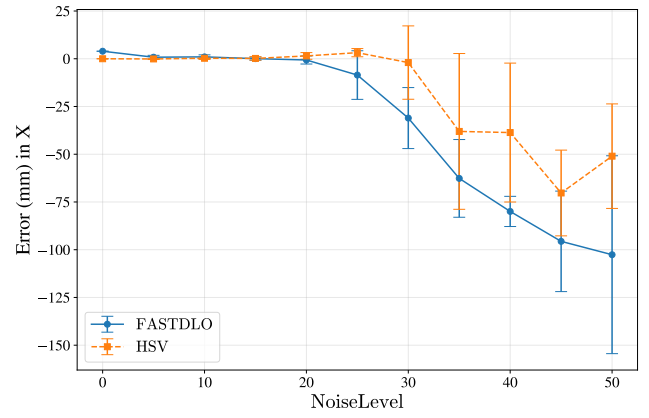
Fig. 8: Skeletonization of FastDLO and HSV masks a) FastDLO mask skeletonization. b) HSV mask skeletonization.

## VI. RESULTS

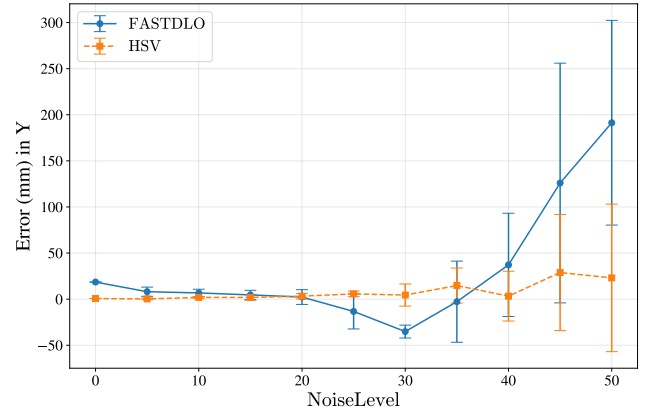
### A. simulation

To evaluate the accuracy and operational limits of the implemented DLO tracking algorithms, they were tested with images with incrementally increasing noise levels. This systematic stress test is visualized using error bar plots (Fig. 6), which capture both the mean estimation error and the stochastic uncertainty—the statistical variability—inherent in the algorithms' outputs under degradation. The robustness analysis specifically targets the spatial localization of Node 2, which represents the designated optimal grasping point for the robotic manipulator. By plotting the  $(x, y)$  coordinate deviation from the ground truth, the algorithm's precision in a task-critical context was quantified.

The results from the noise effect plots for the X and Y coordinates demonstrate a clear relationship between image noise and algorithmic stability:



(a) Effect of noise on X



(b) Effect of noise on Y

Fig. 9: Effect of noise on the results

- **Low Noise Stability (Lower than 30):** For low noise values, the measured X and Y coordinates of Node 2 are highly similar to the expected (ground truth) values. Crucially, the uncertainty in these measurements, as indicated by the length of the error bars, is very low. This suggests that both algorithms perform accurately and consistently under clean conditions.
- **High Noise Instability (Higher than 30):** As the noise level exceeds a threshold of 30, the performance of the algorithms rapidly degrades. The measured X and Y coordinate values begin to diverge significantly from the true values, and the uncertainty (indicated by longer error bars) becomes unstable. This instability suggests that the pixel location of the grasping point cannot be reliably determined as image fidelity decreases.

A *MANOVA* test was conducted for every noise level with the two algorithms as test conditions and the x and y values as dependent variables. Before that, the measurements were tested for normality and heteroscedasticity and compiled in Table II. Even though the normality criterion is not met in all cases, the *MANOVA* test was conducted anyway as it is considered a robust test to some degree of non-normal

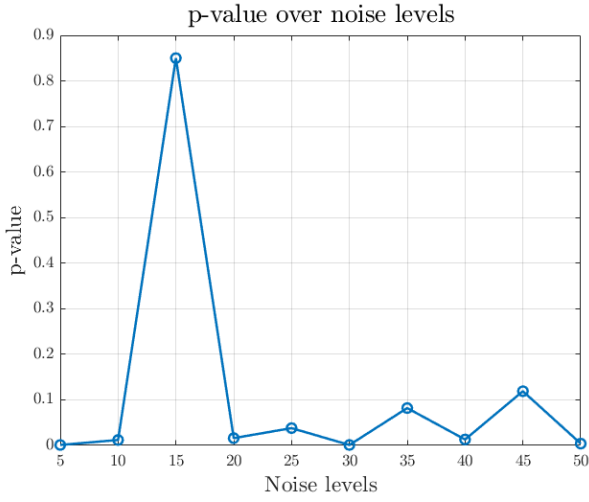


Fig. 10: p-value at different noise levels

conditions. Leech et al. [2005]

Figure 10 shows the statistical significance of the test as the noise levels increase. The difference was statistically significant in some of the noise levels with  $p < 0.05$ . However, no trend was observed as the noise level progressed.

TABLE II: overview of statistical tests

| noise level | p-value | normality | heteroscedasticity |
|-------------|---------|-----------|--------------------|
| 5           | 0,001   | no        | no                 |
| 10          | 0,012   | yes       | yes                |
| 15          | 0,85    | yes       | yes                |
| 20          | 0,016   | no        | yes                |
| 25          | 0,038   | no        | no                 |
| 30          | 0,001   | no        | yes                |
| 35          | 0,082   | no        | yes                |
| 40          | 0,013   | no        | no                 |
| 45          | 0,119   | no        | yes                |
| 50          | 0,004   | no        | yes                |

### B. On real setup

The average and standard deviation of the coordinates of the detected points in the cable (nodes) were computed. For the sake of simplicity, this section focuses on the measurements of the first and second nodes of the cable, as they are the closest to the tip and potential candidates for grasping points. The average location and standard deviation of the detected nodes were used for the analysis of the results.

These measures can provide some information on the uncertainty of the detection.

Tables III and IV summarize this information.

Given the design of the gripper and assuming proper gripper alignment, an error can be tolerated up to 1.7 mm in an axis perpendicular to the cable tested (13 mm). The standard deviation in the x and y axes resulting from the collected data shows that none of the conditions cross these boundaries, proving that all the algorithms can succeed in grasping the cable.

However, the TrackDLO algorithm shows a much lower variance in its measurements. This is particularly important concerning the vertical coordinate, as too much of a deviation could cause the robot to collide with the table.

Fig. 11 - 13 shows the measurements of the first two nodes. For reference, an orange rectangle has been overlaid to show the size of the cable. The figure puts in perspective the size of the uncertainty in comparison with the cable size.

It can also be seen that the uncertainty of the measure is small relative to the diameter of the cable, ensuring the node is detected at the center of the cable and within tolerance of the gripper fingers. This figure assumes that the section of the cable is short enough to be approximated by a cylinder.

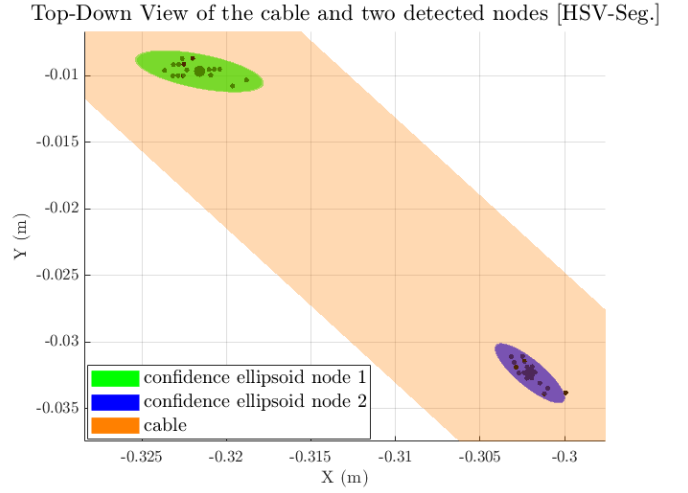


Fig. 11: Top-down view of the cable along two measured points and their confidence ellipsoids in the HSV Segmentation condition

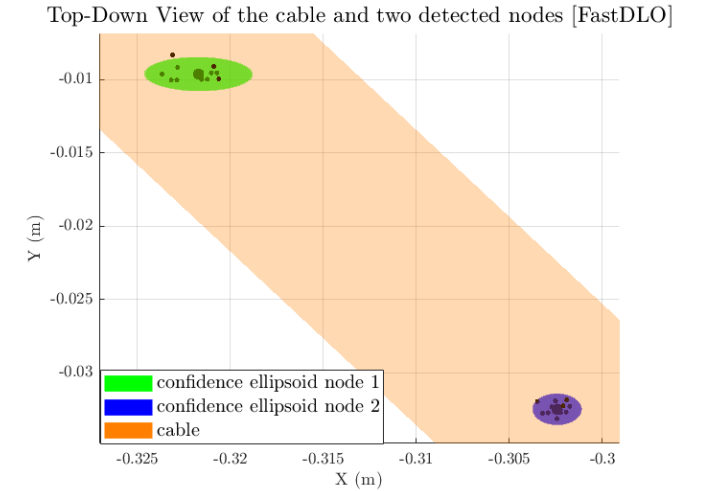


Fig. 12: Top-down view of the cable along two measured points and their confidence ellipsoids in the FastDLO condition

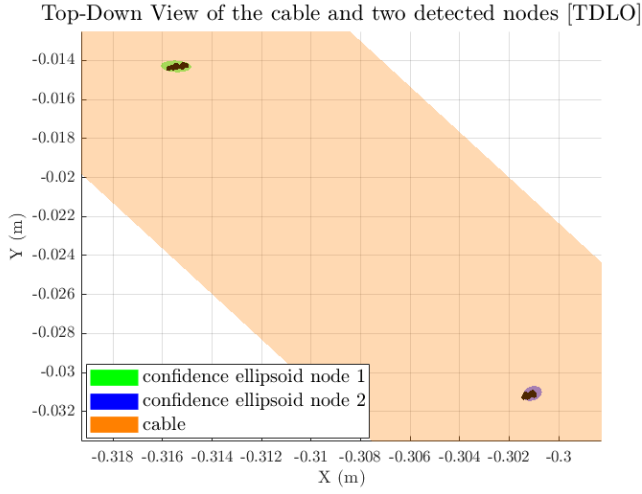


Fig. 13: Top-down view of the cable along two measured points and their confidence ellipsoids in the TrackDLO condition

TABLE III: mean from node 2 measurements

| Algorithm features | Mean of measurements in mm |         |         |
|--------------------|----------------------------|---------|---------|
|                    | X                          | Y       | Z       |
| TrackDLO           | -301.06                    | -310.75 | -446.39 |
| HSV-Seg            | -302.05                    | -323.56 | -444.39 |
| FastDLO            | -302.42                    | -325.01 | -444.42 |

TABLE IV: Standard deviation from node 2 measurements

| Algorithm features | Standard deviation in mm |        |        |
|--------------------|--------------------------|--------|--------|
|                    | X                        | Y      | Z      |
| FastDLO            | 0.4775                   | 0.3884 | 0.3562 |
| HSV-Seg            | 0.7455                   | 0.8097 | 0.3966 |
| TrackDLO           | 0.1378                   | 0.1357 | 0.0554 |

## VII. CONCLUSION

This work is focused mostly on the detection of DLOs and more specifically, cables. However, for this experiment, the thickness of the cable has not been considered as a factor, and therefore, its conclusions should not be expected when utilizing different-sized cables.

More specifically, smaller cables are more difficult to detect and experience larger deformations of their original shape when subjected to gravity, something that can be problematic for insertion tasks.

The algorithms compared in this work can see its performance greatly impacted by lighting conditions. In the data collection for this experiment, relatively constant lighting conditions were assumed due to stable weather and setup. Small changes in lighting can lead to reflections on the cable's surface or changes in the color perceived by the camera that can impact the quality of the detection.

Regarding the statistical tests on the simulation data, it is important to note that the requirements for normality in all groups of the test were not met at all noise levels.

It was earlier introduced in section III. APPROACH that these algorithms automatically detect nodes in the cable without ensuring a fixed distance to the end of the cable. In practice, grabbing the cable too far away from the tip can cause the cable to bend down or be too close to the tip, making the insertion task difficult. Ensuring a proper distance to the end of the tip and quantifying the insertion success rate across different sizes of cables is left for a future study.

### Individual contribution:

- Tilo Wöbber: Camera calibration, Robot programming
- Daniel Aguirre: Simulation and data analysis
- Pol Barrera: Algorithm implementation and data collection

## VIII. ACKNOWLEDGEMENTS

Generative AI usage:

Generative AI was used for rephrasing and language correctness. OpenAI [2025]

## REFERENCES

- Gary Bradski. The opencv library. *Dr. Dobb's Journal of Software Tools*, 2000. URL <https://opencv.org/>.
- Antonio Caporali, Kevin Galassi, Riccardo Zanella, and Gianluca Palli. Fastdlo: Fast deformable linear objects instance segmentation. *IEEE Robotics and Automation Letters*, 7(4): 9075–9082, October 2022.
- Hsin-Hung Chen, Wen-Ning Chuang, and Chia-Chi Wang. Vision-based line detection for underwater inspection of breakwater construction using an ROV. *Ocean Engineering*, 109:20–33, 2015. doi: 10.1016/j.oceaneng.2015.08.012.
- Nancy L. Leech, Kevin C. Barrett, and George A. Morgan. Manova's robustness to violations of normality, 2005. as cited in summary articles.
- Jan Matas, Stephen James, and Andrew J. Davison. Sim-to-real reinforcement learning for deformable object manipulation. In *Proceedings of the 2nd Conference on Robot Learning (CoRL)*, Zürich, Switzerland, 2018.
- OpenAI. Chatgpt (gpt-5.2), 2025. URL <https://chat.openai.com/>.
- Paritosh Parmar. Use of computer vision to detect tangles in tangled objects. *arXiv preprint arXiv:1405.4802*, 2014. URL <https://arxiv.org/abs/1405.4802>.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Jonas Spillmann and Matthias Teschner. Corde: Cosserat rod elements for the dynamic simulation of one-dimensional elastic objects. In *Proceedings of the Eurographics/ACM SIGGRAPH Symposium on Computer Animation*, pages 1–10, 2007.
- R.Y. Tsai and R.K. Lenz. A new technique for fully autonomous and efficient 3d robotics hand/eye calibration.

*IEEE Transactions on Robotics and Automation*, 5(3):345–358, 1989. doi: 10.1109/70.34770.

Jing Xiang, Holger Dinkel, Hao Zhao, Ningyuan Gao, Brian Coltin, Troy Smith, and Timothy Bretl. Trackdlo: Tracking deformable linear objects under occlusion with motion coherence. *IEEE Robotics and Automation Letters*, July 2023. Accepted.